Data Mining for Big Data: Recommender Systems

The utility matrix R contains the ratings of users on different items: r_{xi} is the rating of user x for item i. We denote r_j the vector of the ratings of item i and $r_{x\_}$ the vector of user x ratings. Warning: this matrix may be represented with users in columns or rows depending on the approach (item-item or user-user collaborative filtering).

Only some of the ratings r_{xi} are known. The objective of a recommender system is to compute estimate for unknown r_{xi}.

1 Content Based

Assuming items profiles are given: for Item i, its profile is a vector P_i. For a user x, a profile UP_x can be computed as a weighted average of the items rated by x: $UP_x = \sum_i r_{xi}.P_i$ (the sum is on every item i for which r_{xi} is known).

An estimate of an unknown r_{xi} is based on the cosine similarity between the profile of user x and item i: $UP_x.P_i/||UP_x||.||P_i||$.

2 Item-Item Collaborative Filtering

2.1 Basic CF Estimate

- Given two items i and j, s_{ij} is the centered cosine similarity (or Pearson coefficient) between items i and j (i.e., between vectors r_i and r_j). To compute this similarity, the average rating of item i is subtracted from each known rating in the vector r_i (same for item j and r_j) before computing the cosine similarity. The unknown ratings are taken equal to 0.
- $N(i, x)$ is the set of the nearest neighbors of item i (according to the centered cosine similarity) that are rated by user x.

The estimate for the rating of item i by user x is $r_{xi} = \sum_{j \in N(i,x)} s_{ij}.r_{xj} / \sum_{j \in N(i,x)} |s_{ij}|$

2.2 Global Baseline Estimate

Global baseline estimate for r_{xi} is $b_{xi} = \mu + b_x + b_i$ where μ is the global average of known ratings, b_i is the rating deviation of item i (average rating of item i minus μ) and b_x is the rating deviation of user x.

2.3 Combination of Baseline Estimate and CF

The estimate for the rating of item i by user x is $r_{xi} = b_{xi} + \sum_{j \in N(i,x)} s_{ij}.(r_{xj} - b_{xj}) / \sum_{j \in N(i,x)} |s_{ij}|$

3 Exercice

If an entry of the matrix is empty, it means the rating is unknown. The last row and columns give the average of the ratings on the corresponding row or column. The global average μ of all known ratings in the whole matrix is in the bottom rightmost cell.

	U1	U2	U3	U4	avg.
Item 1	4	5		1	3.3
Item 2		2	3	4	3
Item 3	4		1	1	2
Item 4	5			3	4
Item 5		2	3	1	2
avg.	4.3	3	2.3	2	2.8

Question 1 : Compute the baseline global baseline estimate for the rating $r_{3,1}$ of item 1 by user 3.

Question 2 : What is the centered cosine similarity between items 2 and 3?

Question 3 : Compute the basic CF estimate for the rating $r_{3,1}$ of item 1 by user 3 (assume we take into account the 2 most similar items in this estimation). You can use the following table giving the centered cosine similarity with item 1:

	Item 2	Item 3	Item 4	Item 5
centered cosine similarity with Item 1	-0.96	0.51	0.72	0.56

Question 4 : Same question for the rating of item 5 by user 1.

	Item 1	Item 2	Item 3	Item 4
centered cosine similarity with Item 5	0.56	-0.50	0.0	0.5